Code No: 124CN **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, March - 2022
COMPUTER ORGANIZATION
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) What is instruction format? Discuss about the registers of basic computers.
b) List and explain in detail about various types of interrupts. [8+7]

2. Explain the following addressing modes with examples.
a) Register indirect mode b) Absolute mode c) Direct mode. [5+5+5]

3.a) Why does DMA have priority over the CPU When both request a memory transfer?
b) Compare and contrast between isolated and memory mapped I/O. [8+7]

4.a) Draw the block diagram for asynchronous communication interface.
b) What is handshaking? Discuss with neat diagrams. [8+7]

5 a) Discuss about the virtual memory? Discuss about the mapping of virtual address to memory table.
b) Describe in detail about RAM and ROM organization. [9+6]

6. A two way set associative cache memory uses blocks of four words. The cache can accommodate a total of 2048 words from the main memory. The main memory size is 128 K*32
a) Formulate all pertinent information required to construct the cache memory
b) What is the size of the cache memory [8+7]

7. List and explain all the 16-bit general purpose registers of 8086 microprocessor. [15]

8. Write an 8086-assembly language program for evaluating an arithmetic expression.[15]

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